

74,731

Stockfish-labeled positions

21.3%

best base-model top-1 agreement

Research Question

How much Stockfish move and value knowledge can compact CNN students preserve, and what accuracy-efficiency tradeoffs result from compacting knowledge?

Dataset and Teacher

- Lichess standard rated games, May 2026, streamed directly from `.pgn.zst`.
- Stockfish 18 teacher at depth 10 with MultiPV 5.
- Policy targets are soft distributions over teacher candidate moves from the side-to-move perspective.
- Value targets use White perspective: $\tanh(\text{eval_cp} / 1000)$.
- Final test set: 7,475 positions from 220 games never seen in training or validation.

Corrected Experimental Design

1. Stream games and sample every second ply
2. Label each position with Stockfish MultiPV
3. Split by game, not by position, to prevent leakage
4. Mask illegal moves before policy loss and ranking
5. Train 10 students for 10 epochs; select by validation loss
6. Evaluate once on the held-out test split

Training objective

Loss = policy soft cross-entropy + value MSE

Student Architectures

Two families were tested at five channel/depth settings each:

- **Small CNN**: standard convolutional blocks.
- **Depthwise CNN**: depthwise-separable blocks intended to reduce parameter count.
- Configurations: 16x3, 32x3, 64x3, 32x5, and 64x5.

Architecture Accuracy vs Latency

To

0.215

0.210

SCNN-3

0.205

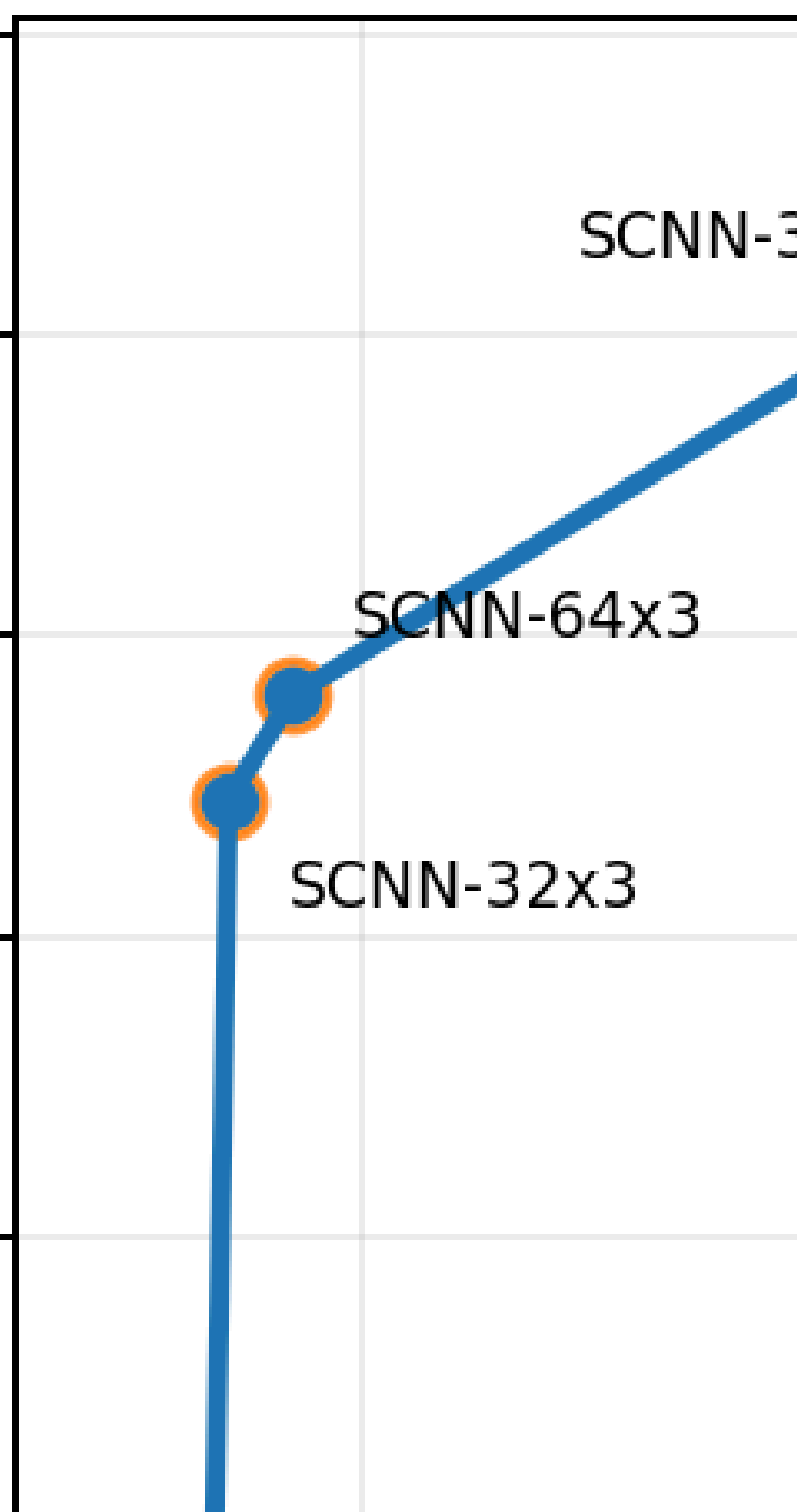
SCNN-64x3

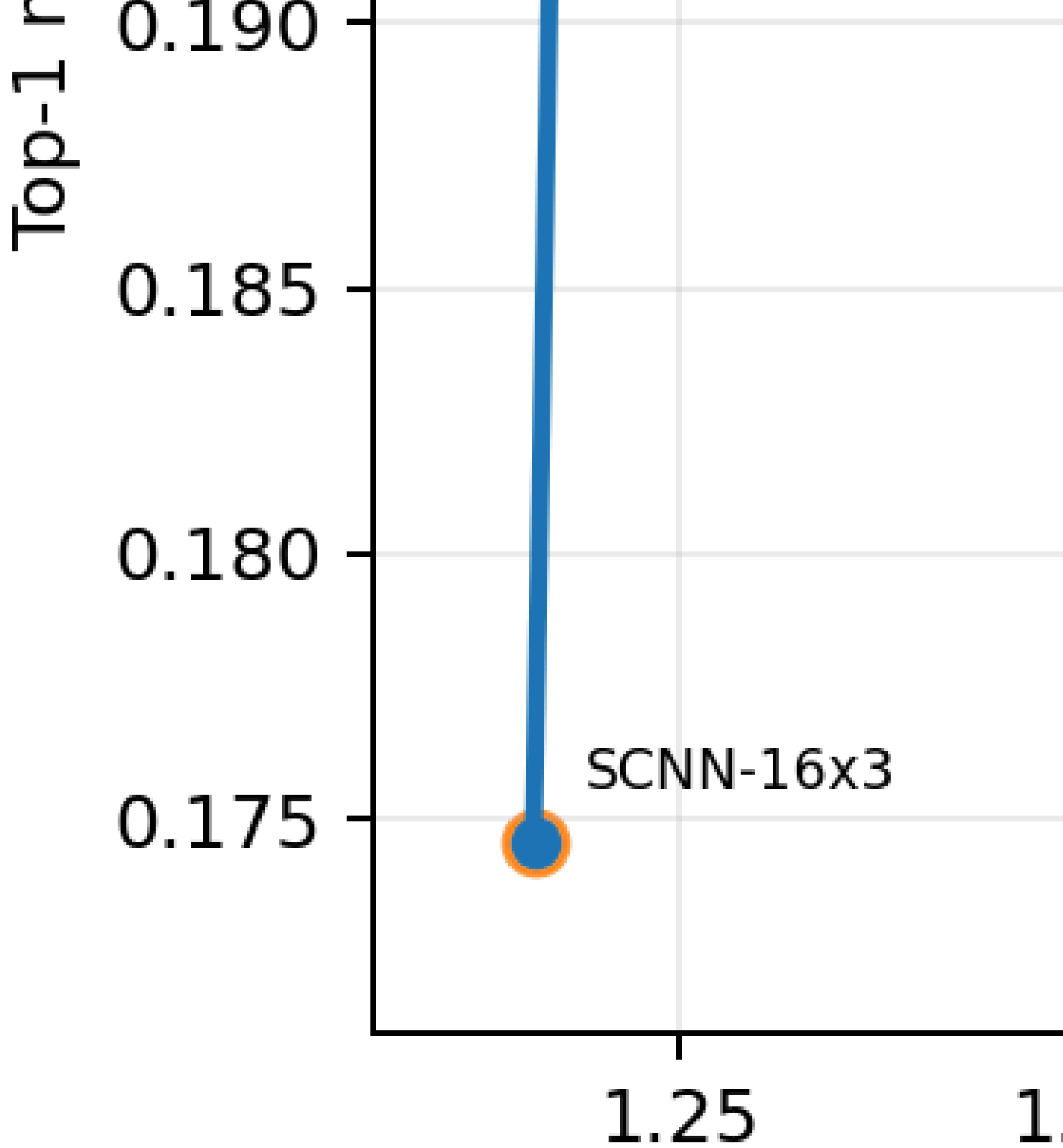
0.200

SCNN-32x3

0.195

move accuracy





Standard small CNNs dominate the tested depthwise variants. SCNN-32x5 gives the best base top-1 agreement; SCNN-32x3 provides the strongest value RMSE.

Held-Out Results

Model

SCNN-32x5

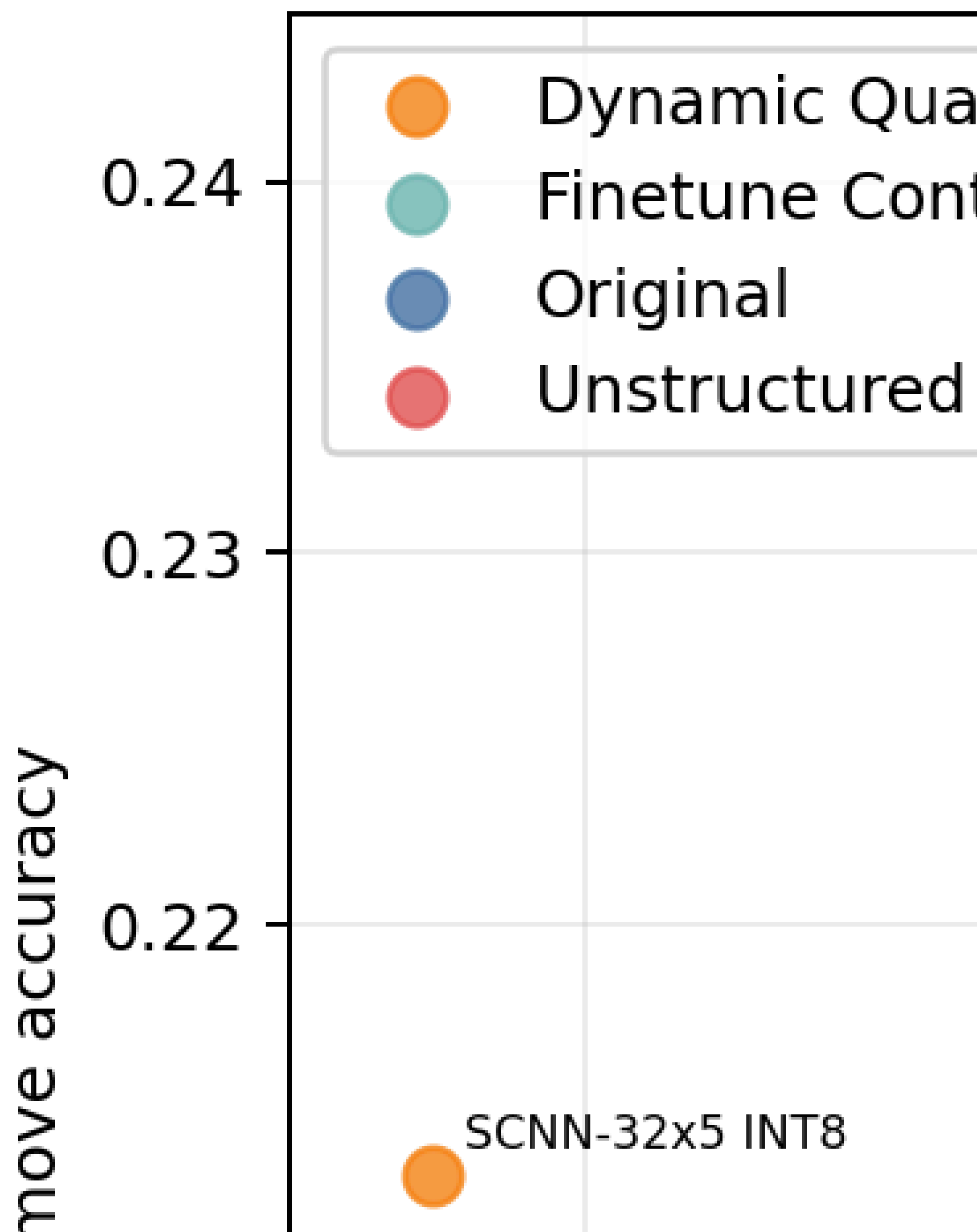
Model

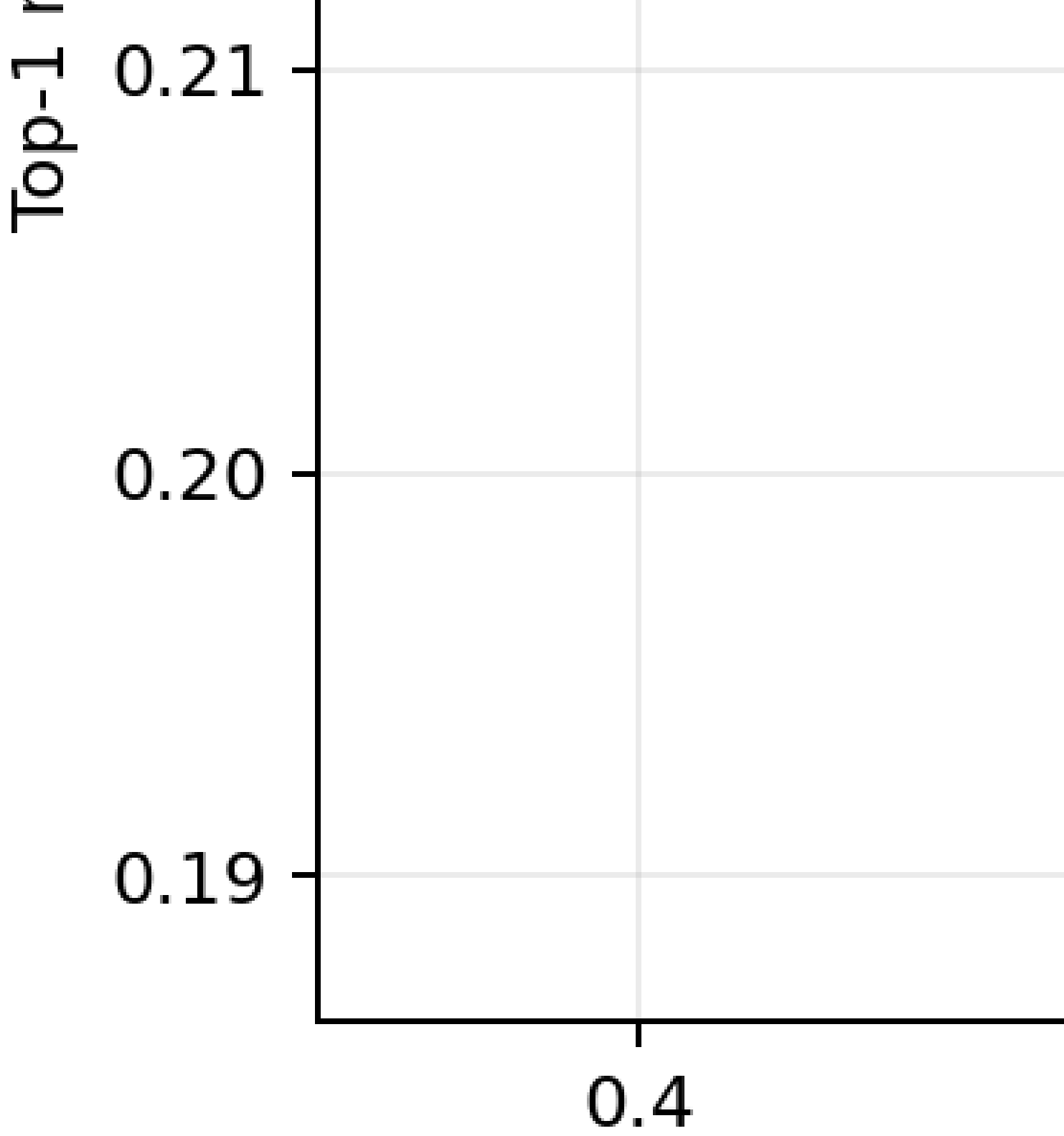
SCNN-64x5

SCNN-64x3

SCNN-32x3

Compression Accuracy vs Size

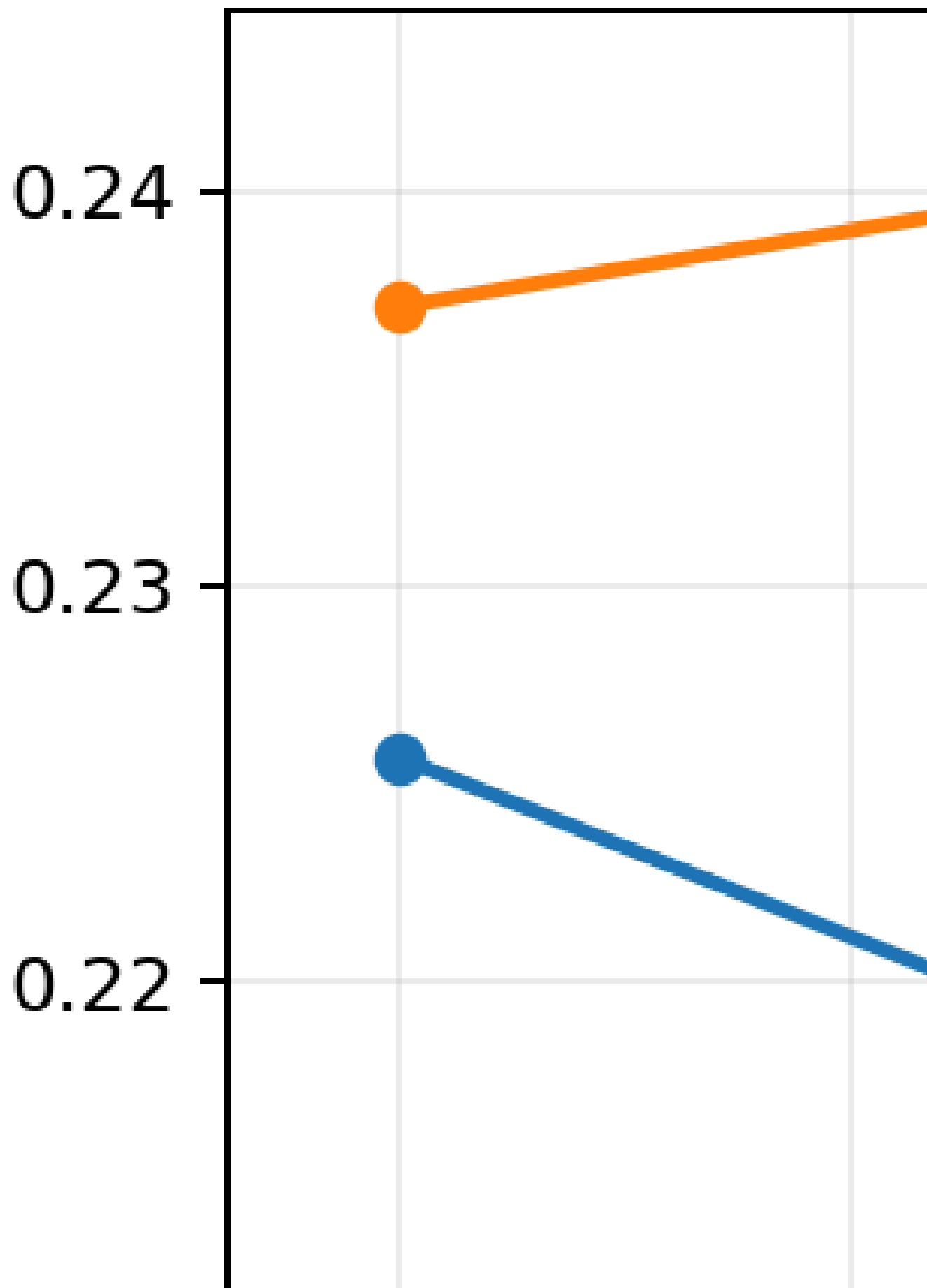


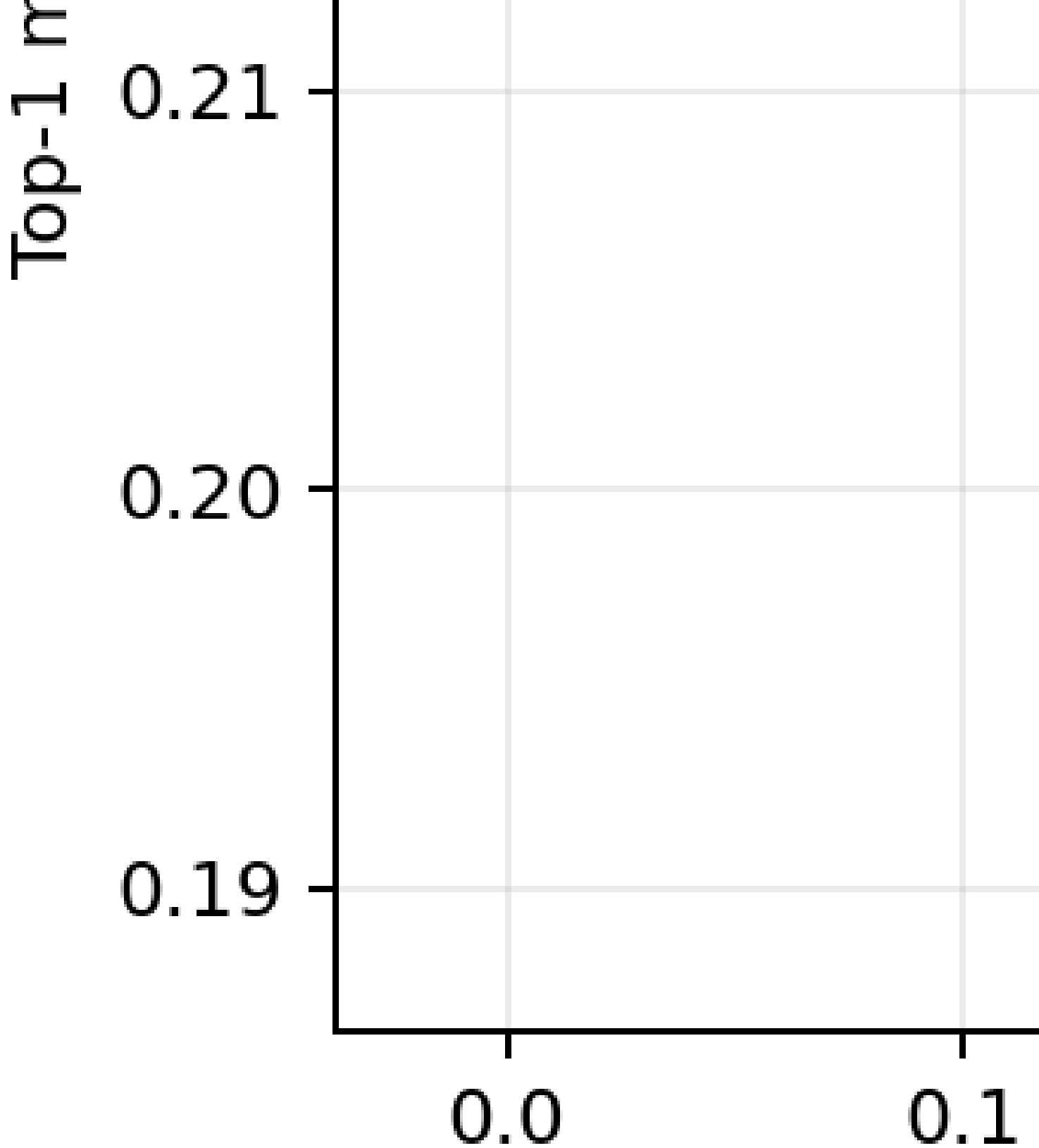


Dynamic INT8 quantization preserves the original top-1 result while reducing checkpoint size. Unstructured pruning leaves dense checkpoint size nearly unchanged.

Fair Pruning Comparison

move accuracy





The zero-sparsity point is an unpruned model given the same two fine-tuning epochs. This control prevents extra training from being mistaken for a pruning benefit.

Key Findings

1

Small CNNs won. Every tested depthwise student was below 19.4% top-1, while four standard CNNs exceeded 20%.

- 2

INT8 is compact, not faster here. Accuracy was unchanged and size fell from 0.72 to 0.35 MB for SCNN-32x5, but measured CPU latency increased.
- 3

Moderate pruning is resilient. SCNN-64x5 retained its fine-tuned accuracy through 50% sparsity; both models declined sharply at 75%.
- 4

Fine-tuning is a major confound. The SCNN-64x5 control improved from 20.6% to 23.7% without pruning.

What Compression Actually Bought

SCNN-32x5
Original
INT8
Fine-tune control
50% pruned
Important: sparsity is not automatically storage compression. A sparse runtime or sparse serialization format is required to convert zero weights into smaller files or faster inference.

Conclusion and Next Steps

For these small chess students, architecture choice mattered more than naive compression. Dynamic quantization offers a clean storage win; moderate pruning preserves accuracy but needs s

- Repeat top configurations across random seeds.
- Use quantization-aware training and static convolution quantization.
- Benchmark sparse kernels and structured channel pruning.
- Evaluate playing strength through fixed-budget engine matches.